

Round 1

1. Distributed API Gateway with Rate Limiting

Tags: Backend · Distributed Systems

Problem: Companies must protect APIs from excessive requests, distribute traffic, and authenticate clients.

MVP:

- API key and JWT authentication
- Request routing and load balancing
- Per-user and per-API rate limiting with Redis counters
- Request logging

Stretch: Circuit breakers and timeouts, usage quotas, admin dashboard for API plans, OpenTelemetry tracing.

Stack: Go/Node.js, Redis, PostgreSQL, Docker, NGINX

Evaluated on: Token bucket vs. sliding window, race conditions, gateway failure handling.

2. Document Processing and Automation SaaS

Tags: Backend · SaaS · Workflow Automation

Problem: Manual processing of invoices, forms, and contracts causes delays and inconsistent data.

MVP:

- Bulk upload of PDFs and images
- OCR and field extraction
- Async job queue
- Human review for uncertain fields
- Export to CSV/JSON

Stretch: Custom templates, DOCX export, multi-tenant access control, retries, duplicate prevention.

Stack: React, TypeScript, FastAPI, Celery/RabbitMQ, PostgreSQL, S3, Docker

Evaluated on: Idempotency, retries, file security, validation.

3. Engineering Productivity and Code Quality Platform

Tags: DevOps · Analytics

Problem: Teams struggle to find delivery bottlenecks from scattered repo and CI data.

MVP:

- GitHub OAuth plus PR, commit, and workflow-run ingestion
- Build failure rate and time-to-merge tracking
- Team-level dashboard with trend charts

Stretch: Flaky test detection, deployment frequency and lead time, incident-to-release linking.

Rules: Team-level reports only; no ranking of individual developers.

Stack: Python/Node.js, PostgreSQL, React, Docker, GitHub Actions

Evaluated on: Metric definitions and avoiding misleading conclusions.

